How Significant Is the Pillar 43 Sky Map?

A Statistical Decomposition of the Göbekli Tepe Archaeoastronomy Claim

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This version supersedes versions 1 and 2. Appendix A lists what changed.

Abstract

Sweatman & Tsikritsis (2017) proposed that animal carvings on Pillar 43 at Göbekli Tepe encode star constellations and a date near 10,950 BC, read from precession of the equinoxes. That date coincides with the proposed Younger Dryas impact event. Sweatman & Gerogiorgis (2025) later stated a combinatorial visual-similarity probability of approximately **1 in 1.4 million**. That figure is the probability of a rank-sum ≤ 18 multiplied by **2/7** (top-row reversal symmetry and a spatial-correlation allowance). It does **not** include a paleoclimatic date term. An exact evaluation of the same product is 7.31×10^{-7} (1 in 1,368,083). We separate that published calculation from an exploratory date test.

Across 13 analyses and two robustness checks we find: (1) the published date does fall within 50 years of the Younger Dryas cluster, but a match of that kind is not rare. **41.8%** of the years from 20,000 to 5,000 BC lie within $\pm$250 years of at least one catalogued event, **14/52 (26.9%)** of constellation × cardinal-day center dates match one, and **30/52 (57.7%)** of the full constellation windows overlap one. Keeping only the events that predate the monument leaves coverage at **33.3%**; (2) within Sweatman's self-scored ranks the visual similarity is unusual ($P = 2.558 \times 10^{-6}$, about 1 in 390,881), but the signal is concentrated in three of six mappings (**1 in 85,184** if those three animals are pre-specified; **1 in 4,484** if the three best ranks are chosen after seeing all six), while the other three remain conventionally significant and much weaker (1 in 187); (3) exploratory date × visual products depend on analyst choices. Across 32 primary paths they span a factor of about **11** (1 in 6.8 million to 1 in 635 thousand), and about **49** once animals are dropped after the ranks are seen. The statistical case depends on three self-scored visual rankings that have not been blindly replicated. A blind ranking experiment would be the decisive test. Code and data are in the repository.

Keywords: archaeoastronomy, Göbekli Tepe, statistical significance, look-elsewhere effect, precession, multiverse analysis

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1. Introduction

Göbekli Tepe, a monumental Neolithic site in southeastern Turkey, has produced some of the most debated archaeological interpretations of the last decade. Among them, the proposal by Sweatman and Tsikritsis (2017) stands out for its specificity: the animal carvings on Pillar 43 in Enclosure D represent star constellations, and the scene encodes the date of the summer solstice at approximately 10,950 BC via precession of the equinoxes. This date coincides with the proposed Younger Dryas impact event (Kennett et al., 2015; see also Holliday et al., 2023, for a critical review of the Younger Dryas Impact Hypothesis), leading the authors to interpret the pillar as a memorial to a cosmic catastrophe. The claim was published in *Mediterranean Archaeology and Archaeometry* and has since been extended to European Palaeolithic cave art (Sweatman & Coombs, 2019), a lunisolar calendar embedded in the same pillar's geometric symbols (Sweatman, 2024), and a further probabilistic analysis (Sweatman & Gerogiorgis, 2025).

The excavation team responded with a qualitative rebuttal (Notroff et al., 2017), raising objections about pillar repositioning during multi-phase rebuilding, a 700–1,000 year gap between the proposed date and the site's radiocarbon chronology, and the assumption that Neolithic hunters recognized Greek-derived constellations, an assumption the rebuttal treats as highly arbitrary. These are serious concerns. Neither the rebuttal nor later work published the quantitative decomposition attempted here: whether the stated statistical evidence is as strong as claimed, or whether it arises from a dense event record, subjective visual matching, and researcher degrees of freedom.

This paper evaluates Sweatman's **visual-similarity** calculation (the rank-sum and the published **2/7** multiplier) and, separately, a **date-coincidence** component under paleoclimatic look-elsewhere models. Joint products of those components are reported as exploratory and are not identified with the 1.4-million headline.

The date component is readily obtained under the null used here. The visual component is concentrated in three of six mappings, and that concentration is weaker once the three mappings are recognized as the best of six rather than as a pre-specified trio. The paper proceeds as follows. Section 2 is background. Section 3 is methods. Section 4 is the 13 analyses. Section 5 is implications, limitations, and the blind-test proposal.

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2. Background

2.1 Göbekli Tepe and Pillar 43

Göbekli Tepe is located near Şanlıurfa in southeastern Turkey, approximately 15 km northeast of the city centre. Excavations led by Klaus Schmidt (DAI) from 1995 until his death in 2014, and continued by the Şanlıurfa Museum and DAI, have uncovered a sequence of roughly circular enclosures formed by T-shaped megalithic pillars and rough stone walls (Schmidt, 2003; Dietrich et al., 2012). Enclosure D plaster and associated contexts yield calibrated radiocarbon ranges in the mid-to-late 10th millennium cal BC (Dietrich et al., 2013; e.g. sample KIA-44149: 9745–9314 cal BC). Treating these as a single Gaussian such as "9,530 $\pm$ 215 BCE" would overstate precision and would present a site-wide estimate the excavation report does not make. The chronological gap relative to ~10,950 BC is left as qualitative context from the excavation team.

Pillar 43 is embedded in the northwestern portion of Enclosure D's perimeter wall. Its western broadside carries an elaborate relief: a large vulture or eagle with a circular disc on its wing, a scorpion below, a canid to the left, a tall bird grasping a fish or snake to the right, a partially ob-

scured swimming bird at the bottom, and several smaller figures. The narrow faces carry further carvings, including snakes and a spider. A headless human figure appears at the base.

The excavation team reads the scene in the context of Pre-Pottery Neolithic mortuary ritual and excarnation (Peters & Schmidt, 2004; Notroff et al., 2016). The vulture and headless man are referred to a documented tradition of defleshing by raptors. The disc is sometimes read as a severed head.

2.2 Sweatman's astronomical interpretation

Sweatman and Tsikritsis (2017) identified animal symbols on Pillar 43 with star constellations, using the modern Western set as drawn in Stellarium. The mappings used here are in Table 1.

Table 1. Animal–constellation mappings for Pillar 43. Ranks are those published by Sweatman & Gerogiorgis (2025) against a pool of **44** constellations: Ptolemy's 48 minus four southern figures they treat as not visible from the site at the relevant epoch (Canis Major, Lepus, Eridanus, and Argo Navis). That pool is not a census of every modern constellation above the horizon at ~37.2°N. Ranks 4, 5, and 6 are published as upper bounds ("no worse than"), so 18 is an upper bound on the sum. The vulture is the date anchor and is excluded from the rank-sum. The duck/goose is excluded as partly obscured. The authors call the Scorpius and Lupus ranks objective and note some uncertainty in the Ophiuchus rank. On archaeological grounds the canid → Lupus identification is the one marked **tentative** here (Section 5.2): the carving is a canid, and the wolf/dog species call is not independently settled.

Animal	Constellation	Rank (of 44)	Role
Vulture/eagle	Sagittarius	—	Anchor (excluded)
Scorpion	Scorpius	1	Tested
Wolf/dog	Lupus	1	Tested (tentative)
Tall bird + snake	Ophiuchus	1	Tested (authors note uncertainty)
Bending bird	Pisces	4	Tested
Ibex/quadruped	Gemini	5	Tested
Frog/bear	Virgo	6	Tested
Duck/goose	Libra	—	Excluded (obscured)

The date is derived in five steps. The disc on the vulture's wing is identified as the Sun, by analogy with a disc beside a crescent on Pillar 18. The vulture is Sagittarius, specifically the "teapot" asterism. The Sun in Sagittarius is read as the summer solstice, because a winter solstice in Sagittarius falls near 2000 AD. With a precession period of about 25,772 years, the summer-solstice Sun lay in Sagittarius near 10,950 $\pm$ 250 BC (Sweatman & Gerogiorgis, 2025, quote the same date as 10,950 $\pm$ 200 BCE). Three "handbag" symbols are then read as the other cardinal stations: **Virgo for the spring equinox, Gemini for the winter solstice, and Pisces for the autumn equinox**.

The published date is not the midpoint of Sagittarius. It comes from where the disc sits against the constellation. On the boundaries used here the solstice Sun at 10,950 BC is at 270.9°, about 4° inside the Ophiuchus edge of Sagittarius. The midpoint of the constellation corresponds to 11,835 BC, 885 years earlier. Section 3.4 keeps the two apart.

2.3 The probability calculation and its revisions

The published figure has moved more than once in eight years. The path is **not** a sequence of revisions each reducing the claimed significance.

- Sweatman & Tsikritsis (2017) reported approximately **1 in 100 million**. That calculation ranked Göbekli Tepe animal symbols against a small constellation pool and counted favorable combinations. It was **not** the later summed-ranks statistic.
- Sweatman & Coombs (2019) reported approximately **1 in 140 million** for the Vulture Stone match. Moving from 100 million to 140 million **increases** the claimed rarity. Appendix A of that paper multiplies by **13** after treating the scorpion = Scorpius identification as given, and then applies a separate positional factor. The 2019 statistic is still a symbol-combination count, not the 2025 rank-sum.
- Sweatman & Gerogiorgis (2025), a submitted manuscript posted by the authors and not a peer-reviewed journal article, exclude the vulture from the test, compare six animals with 44 constellations, and state a headline of about **1 in 1.4 million**. In their own words the probability is $p \times 2/7$, where p is the probability of rolling a score of 18 or less on six 44-sided dice, the factor 2 is the symmetry of the top row (the order can be reversed and still read as the same date), and division by 7 is the spatial correlation carried forward from 2019. There is no paleoclimatic date in that product. Using a more detailed Greek stick-figure set, their summed score rises from 18 to 26 and the same headline moves to about **1 in 110,000**.

Our exact rank-sum is $p = 2.558 \times 10^{-6}$ (1 in 390,881). Multiplied by 2/7 it is 7.31×10^{-7} (1 in 1,368,083), the same order as the stated 1.4 million. The same formula with a sum of 26 gives 1 in 110,312, which matches their second figure. Sweatman's round numbers come from an online calculator applied to that product, not from a date term.

An exploratory date × visual product can land near the same number by accident. The strict anchor rate in Section 4.1 is $14/52 \approx 0.27$, close to $2/7 \approx 0.286$, so rank-sum × anchor rate (1 in 1,451,844) resembles rank-sum × 2/7. The likeness is a shared rank-sum times a factor near 0.3. It is not evidence that the 1.4 million calculation contains a date.

2.4 The Notroff et al. rebuttal

Notroff et al. (2017) object that Enclosure D's pillars are not all in their original positions, that 10,950 BC is 700–1,000 years older than the oldest radiocarbon dates associated with Enclosure D, that recognizing the Greek constellations in Neolithic art is highly arbitrary, and that the argument selects a few pillars from a site with many animal reliefs.

Sweatman and Tsikritsis (2017b) answered that a probability should outrank an opinion, and that the radiocarbon date dates wall mortar rather than the carving. Neither side published the null models in Section 3.

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3. Methods

3.1 Precession engine

The ecliptic longitude of the vernal equinox is shifted linearly at $360°/25{,}772 \approx 0.01397°$ per year. For a year BC and a cardinal day, the Sun's J2000 ecliptic longitude is compared with the constellation intervals in the script. Two internal checks, which are **not** a Stellarium validation, still pass: at 10,950 BC the summer-solstice Sun is at 270.89°, inside Sagittarius (266.5°–300.0°); at 3,000 BC the vernal equinox is at 69.84°, inside Taurus (53.5°–90.4°).

The Pisces/Aries junction is **28.687°**, the intersection of the IAU (B1875) boundary with the J2000 ecliptic (Asher's reduction). The other twelve junctions are a one-decimal table. A residual table against Stellarium or a full boundary refresh was **not run**. The linear engine is treated as adequate for windows of a few centuries, with about 1° of error over 13,000 years (~70 years).

`constellation_date_range()` returns one short interval centered on the constellation midpoint, with length $\Delta\lambda$ divided by the precession rate. The script refuses to continue if any of the 52 windows disagrees with its angular width by more than a year.

3.2 Paleoclimatic event catalog

The catalog is 14 analyst-defined clusters between 20,000 and 5,000 BC (Table 2). Sources are ice-core chronologies (GICC05), speleothems, radiocarbon-dated geological events, and marine cores. Three pairs of centers are less than 500 years apart and are kept as separate clusters:

- YD termination and PBO / Bond 8 / MWP-1B, 350 years
- the mammoth placeholder and Bond 6, 250 years
- Mazama and the late cluster, 177 years

The 14 clusters are scenarios, not a unique partition of the deglacial record. Each cluster carries a half-width as a dating-uncertainty label. **Primary matches do not use that half-width.** They use a uniform tolerance ($\pm$100, 200, 250, or 300 years in the multiverse; the anchor grid also reports $\pm$500). The half-width is used to build the precision-filtered subset and to scale the catalog-sweep perturbations.

Table 2. Event clusters.

Cluster	Date (BC)	Half-width (yr)	Basis
LGM peak	19,050	1,000	Clark et al. (2009)
Bonneville Flood	15,550	500	O'Connor (1993)
Heinrich Event 1	14,850	500	Hemming (2004), Table 4: 16,800 calendar years (14,200 ^{14}C yr + 2,600), placed with the 1950 BP convention
Bølling-Allerød + MWP-1A	12,700	200	Deschamps et al. (2012)
Older Dryas	12,075	200	Rasmussen et al. (2014)
YD cluster	10,900	250	Reinig et al. (2021); Kennett et al. (2015)
YD termination	9,750	100	Cheng et al. (2020)
PBO / Bond 8 / MWP-1B	9,400	200	Rasmussen et al. (2007)
Bond 7	8,350	500	Bond et al. (1997)
Mammoth extinction (N. Asia)	7,700	300	**Not a calibrated age.** Kuzmin & Orlova (2004): mainland extinction ca. 9700 ^{14}C BP; Wrangel survival ca. 7700–3700 ^{14}C BP. Kept as a placeholder and tested in robustness check B
Bond 6	7,450	500	Bond et al. (1997)
8.2 ka cluster	6,300	300	Barber et al. (1999)
Mazama VEI 7	5,677	150	Zdanowicz et al. (1999): 7627 $\pm$ 150 cal yr BP
Late cluster	5,500	400	Ryan et al. (1997)

The **precision-filtered** subset keeps clusters with half-width $\leq$ 250 years. In this catalog that keeps six clusters (Bølling-Allerød + MWP-1A, Older Dryas, the YD cluster, YD termination, PBO / Bond 8 / MWP-1B, and Mazama). Heinrich 1, with half-width 500 years, is excluded. The cut is dating precision, not event size. A precision-filtered event is still matched at the scenario tolerance, which can be wider than its half-width.

3.3 Visual similarity model

Each of the six tested animals is ranked independently against the 44-constellation pool. The published ranks are [1, 1, 1, 4, 5, 6], sum 18. $P(S \leq 18)$ for six independent uniform draws from $\{1, \ldots, 44\}$ is computed by inclusion-exclusion. Three-animal subgroups are checked by enumeration.

The published multiplier is evaluated directly: $P_{2025} = P(S \leq 18) \times (2/7)$. That quantity is compared with the exploratory date × visual products. It is not replaced by them.

3.4 Null models

Anchor null (date, exploratory). A random assignment picks one of 13 ecliptic constellations and one of four cardinal days (52 combinations), computes the precession date, and asks whether it meets the catalog. The strict variant uses the constellation center. The broad variant asks whether the constellation's date interval overlaps any event window at the stated tolerance. A third variant, the position null, draws a date uniformly inside the constellation's interval and averages over the 52 combinations. Only **30 of the 52** center dates fall inside 20,000–5,000 BC. The other 22 can never match, because the catalog stops at the window edge, so the strict rate is reported both over all 52 anchors and over the 30 that land in the window. Coverage (uniform over years) is a further baseline on a different sampling domain. Statements are conditioned on the domain.

Observed reading. The observation set against these nulls is the date Sweatman publishes, 10,950 BC. It is not the Sagittarius midpoint (11,835 BC), which is only what the strict null draws for that constellation.

Similarity null. Each animal draws a rank uniformly from $\{1, \ldots, 44\}$. $P(S \leq 18)$ is the visual p-value under that null.

Joint exploratory product. $P(\text{joint}) = P(\text{anchor match}) \times P(S \leq 18)$, assuming independence. This is not the 2025 headline. If the ranks were influenced by the target date, the joint probability is higher (less rare).

Multiverse. Paths cross anchor model (center vs window), tolerance (100, 200, 250, 300 yr), catalog (all 14 vs precision-filtered), cardinal scope (summer solstice only vs all four), and animal inclusion (all 6, or the best 5, 4, or 3). There are 128 paths, of which 32 keep all six animals (the primary set).

Validity filter. A path is valid only when the published date is a hit under that path's catalog and tolerance. The published date is 50 years from the Younger Dryas cluster, which is in both catalogs, and the smallest tolerance in the grid is $\pm$100 years, so all 128 paths are valid. Zero-probability paths are counted rather than dropped, and this run has none. The reference path is summer solstice, all events, $\pm$250 years, center anchor, all six animals. Its joint is an exploratory product, not $p \times 2/7$.

Robustness checks. Check A removes every cluster later than 9,500 BC, a round cut-off inside Enclosure D's calibrated range, and shortens the window to 20,000–9,500 BC. A memorial cannot record an event that came after it. Check B removes the mammoth cluster, and separately moves it to about 9,150 BCE, an approximate IntCal20 age for ca. 9700 ^{14}C BP assuming a 1σ of 60–100 ^{14}C years (the source gives no error term).

3.5 Software and reproducibility

One Python script, `gobekli_statistical_test.py`, uses only numpy and matplotlib. Constellation boundaries, the event catalog, and the ranks are in the script. The committed out-

puts are `output/run_log.txt`, `output/results.json`, and `output/fig1_timeline.png` through `output/fig5_catalog_sweep.png`. Seed 42 fixes the catalog sweep, the circular shift, the rank-sum histogram, and the identity permutation. Every number in Section 4 is in those two output files.

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4. Results

4.1 The date coincidence is not rare

Anchor test (Experiment 1). Of 52 constellation × cardinal-day combinations at ±250 years, **14 (26.9%)** match on the center date and **30 (57.7%)** match on the window (Figure 4). Among the 30 combinations whose center lies inside 20,000–5,000 BC the strict rate is **14/30 (46.7%)**. Under the position null, a date drawn at random inside a random constellation window matches **23.9%** of the time. At ±100 years the strict rate is **7/52 (13.5%)**. The 14 strict matches at ±250 years cover **10** of the 13 ecliptic constellations and all four cardinal days, from the Bonneville Flood to the late cluster.

The observed reading. Figure 1 shows summer-solstice windows against the catalog. Sagittarius at the summer solstice runs from **10,635 to 13,034 BC**, center **11,835 BC**. The published date, 10,950 BC, is 885 years from that center, near the Ophiuchus edge of the window, and 50 years from the Younger Dryas cluster (10,900 BC). It is a match at every tolerance used here. The date test asks how often a match of that kind turns up, not whether this one is real.

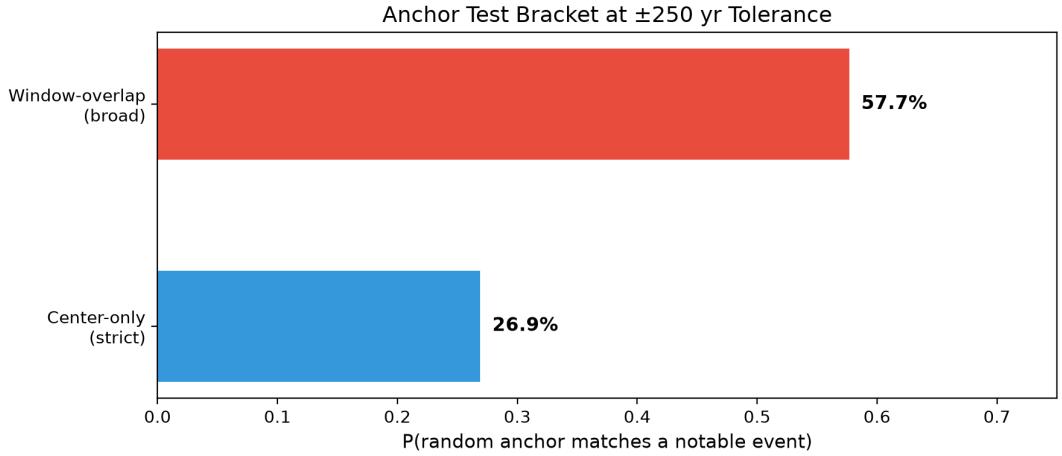


Figure 4. Anchor test at ±250 years: center-date matches and window matches.

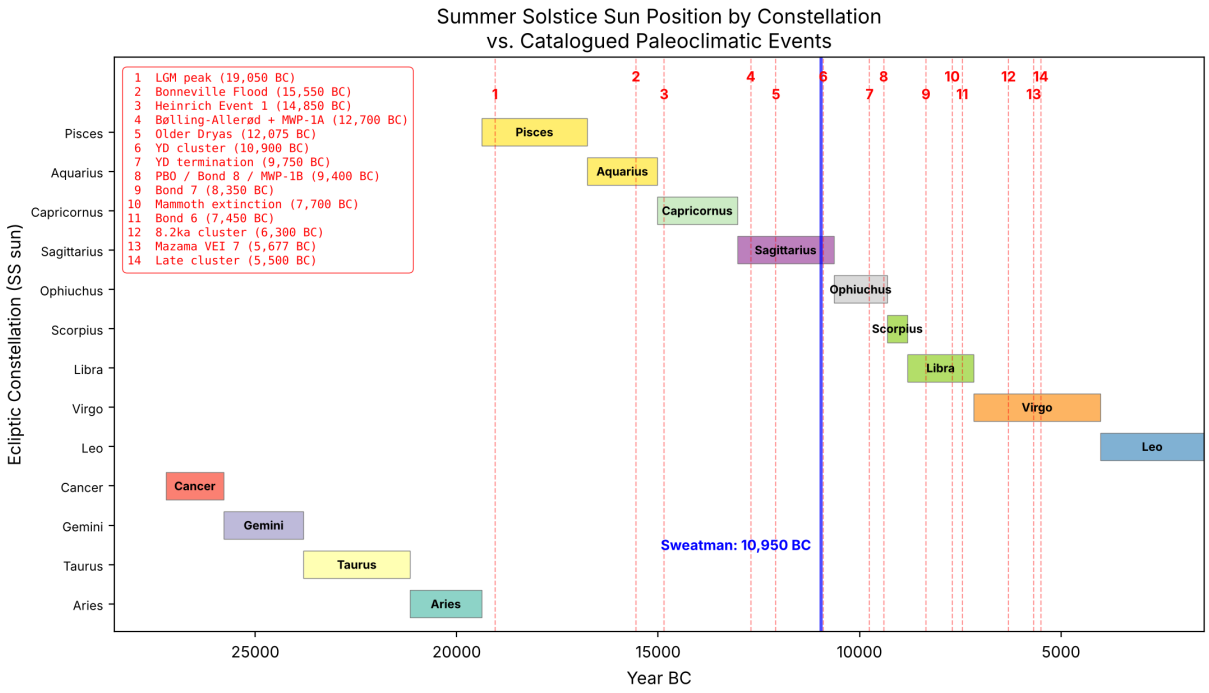


Figure 1. Summer-solstice windows against the catalog. Red numbers mark the 14 event clusters. The blue line is the published date.

Coverage (Experiment 9). **41.8%** of the 15,000 years from 20,000 to 5,000 BC lies within $\pm$250 years of at least one catalogued event. At $\pm$100 years coverage is 18.5%; at $\pm$500 years it is 69.2%.

Leave-one-event-out (Experiment 7). The $\pm$250 year center rate is 26.9% (14/52). Removing the Older Dryas drops it to **21.2%** (11/52), a **5.8** percentage-point fall and the largest single-cluster effect. Removing the Younger Dryas cluster leaves all 14 hits in place, because none of the 52 center dates lies within 250 years of it. No single cluster carries the null rate.

Circular shift (Experiment 10). Shifting the whole catalog inside 20,000–5,000 BC, wrapping modulo 15,000 years, preserves the number of events and their spacing. Across 1,000 shifts, **37.6%** produce a $\pm$250 year center rate at least as high as the 26.9% baseline (shifted mean 24.1%). Event density does most of the work; the real placement is not an unusually favorable alignment under this shift model.

Catalog sweep (Experiment 5). Five hundred catalogs drop 0–3 events and shift the remaining dates by a normal draw with standard deviation half the cluster half-width. The $\pm$250 year center rate runs from **9.6% to 38.5%** (mean **25.7%**, median 25.0%; Figure 5). The lower end remains near 10%.

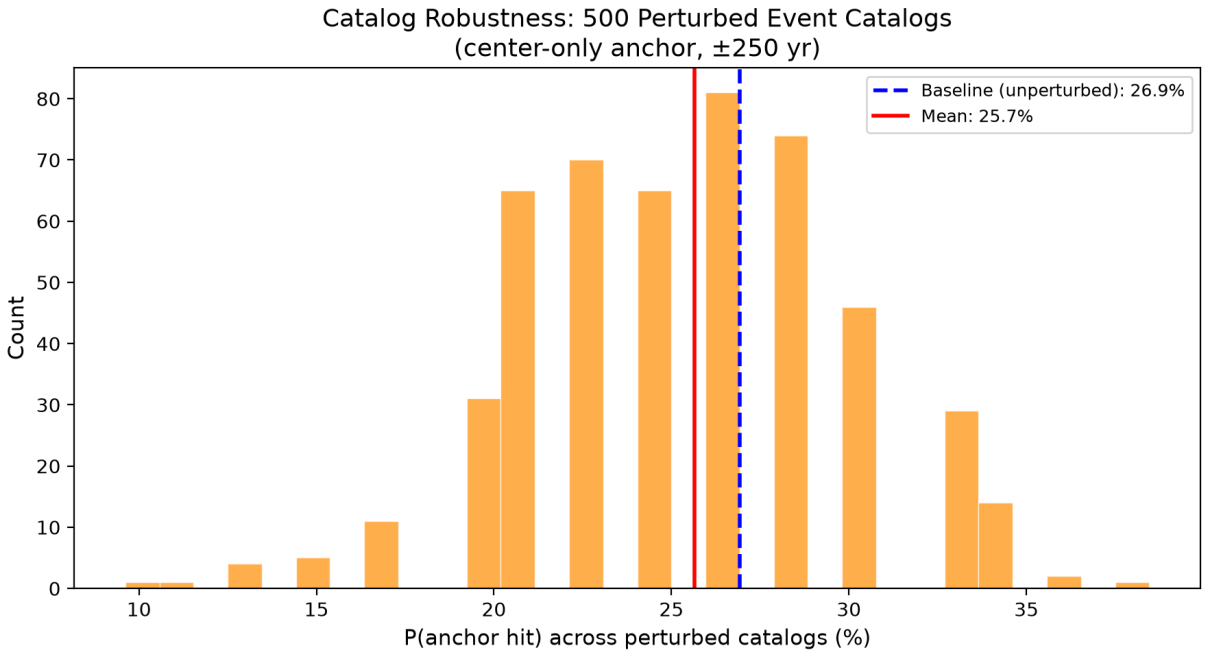


Figure 5. Center-date hit rate across 500 perturbed event catalogs ($\pm$250 years).

Events that predate the monument (robustness check A). Seven of the 14 clusters, PBO / Bond 8 / MWP-1B and everything after it, are later than 9,500 BC and so later than the monument. With those removed and the window cut to 20,000–9,500 BC, coverage at $\pm$250 years is **33.3%**. The strict rate is **7/52 (13.5%)**, or **7/19 (36.8%)** among centers that land in the shorter window, and the broad rate is **21/52 (40.4%)**. At $\pm$100 years coverage is 13.3%. The baseline falls, as it should, and a date match is still about a one-in-three event.

The mammoth cluster (robustness check B). Removing the cluster leaves the strict rate at 14/52 and the broad rate at 30/52, with coverage 40.2%. Moving it to 9,150 BCE gives 15/52, 30/52, and 41.8%. The placeholder does not carry the result.

4.2 The visual similarity is unusual and concentrated

Exact rank-sum (Experiment 2). $P(S \leq 18) = 2.558 \times 10^{-6}$, 1 in 390,881. The null expectation is 135 $\pm$ 31. The observed sum 18 is 3.76 standard deviations below that mean (Figure 3). Because ranks 4 to 6 are published as upper bounds, this is an upper bound on the probability the published ranks imply. These rank-sum figures do not depend on the event catalog.

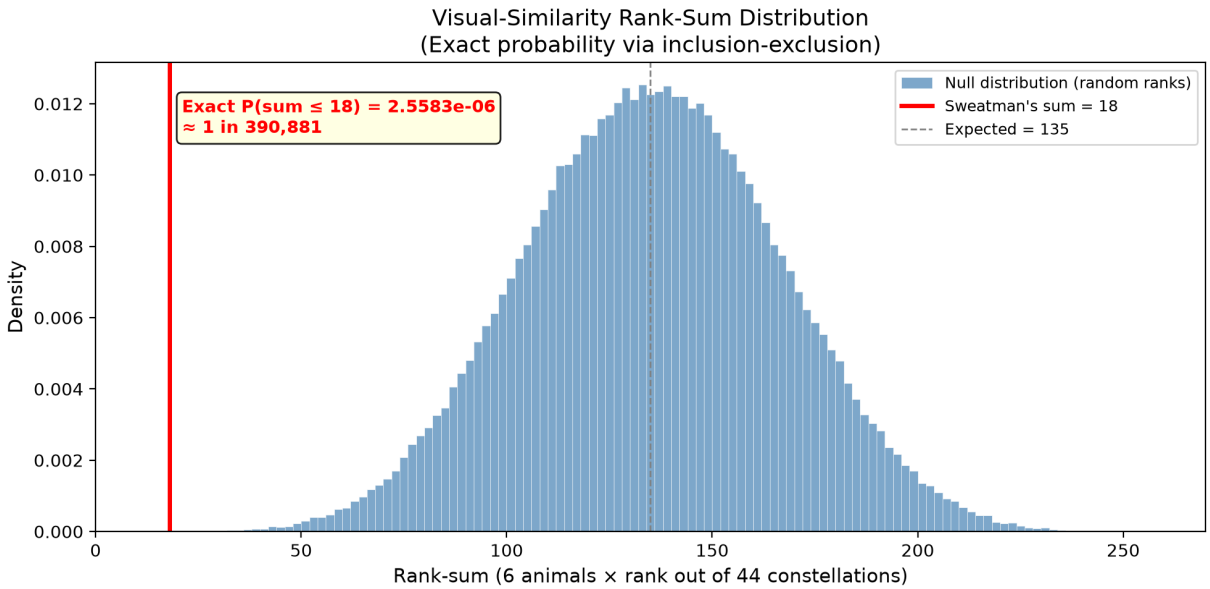


Figure 3. Rank-sum null distribution. The observed sum is 18.

The 2/7 factor. $2.558 \times 10^{-6} \times 2/7 = 7.309 \times 10^{-7}$, 1 in 1,368,083. That is the published headline's order of magnitude, with no date term. With the Greek stick-figure sum of 26 the same formula gives 9.07×10^{-6}, 1 in 110,312, the published second figure. A date look-elsewhere correction does not "explain" the 1.4 million figure from inside the 2025 formula. It answers a different question.

Decomposition (Experiment 6). Sorting the six ranks and splitting them into [1, 1, 1] and [4, 5, 6] is what the code does. For three **pre-specified** animals, $P(\text{all three rank } 1) = 1/44^3 = 1/85{,}184$. For the operation actually performed, keeping the best three of six, the null probability of three or more rank-1 outcomes is

$$\sum_{k=3}^{6} \binom{6}{k} \left(\frac{1}{44}\right)^{k} \left(\frac{43}{44}\right)^{6-k} = 2.230 \times 10^{-4},$$

which is **1 in 4,484**. The bottom three have $P(S \leq 15) = 5.34 \times 10^{-3}$ (1 in 187). The product of the pre-specified top-three probability and the bottom-three probability is 6.27×10^{-8}, which is **not** $P(S \leq 18)$. Table 3 is a diagnostic split, not a factorization.

Table 3. Rank-sum decomposition. Inclusion-exclusion, checked by enumeration for the groups of three.

Group	Ranks	Sum	P	≈ 1 in
Top 3, pre-specified (scorpion, canid, Ophiuchus)	[1, 1, 1]	3	1.17×10^{-5}	85,184
Top 3, sort-then-pick	at least three 1s among six	—	2.23×10^{-4}	4,484
Bottom 3 (Pisces, Gemini, Virgo)	[4, 5, 6]	15	5.34×10^{-3}	187
All 6	[1, 1, 1, 4, 5, 6]	18	2.56×10^{-6}	390,881

Leave-one-animal-out (Experiment 11). Dropping any one rank-1 animal multiplies the rank-sum probability by 14.7 (Table 4). The three rank-1 animals tie. Dropping the bending bird (rank 4) weakens the result by 4.7×, the ibex (rank 5) by 3.1×, and the frog/bear (rank 6) by 1.9×.

Table 4. Leave-one-animal-out.

Dropped animal	Rank	Remaining P	≈ 1 in	Weakening
Scorpion	1	3.75×10^{-5}	26,651	14.7×
Wolf/dog	1	3.75×10^{-5}	26,651	14.7×
Tall bird + snake	1	3.75×10^{-5}	26,651	14.7×
Bending bird	4	1.21×10^{-5}	82,376	4.7×
Ibex/quadruped	5	7.80×10^{-6}	128,140	3.1×
Frog/bear	6	4.80×10^{-6}	208,228	1.9×

Identity permutation (Experiment 13). Conditional on a rank pattern with three rank-1 slots, the probability that one specific trio occupies all three is $1/\binom{6}{3} = 1/20 = 5\%$. That is a complement to the decomposition, not a separate discovery.

4.3 The exploratory joint depends on the path

Joint product (Experiment 3). At $\pm$250 years the strict joint is 6.89×10^{-7} (1 in 1,451,844) and the broad joint is 1.48×10^{-6} (1 in 677,527). Neither number corrects the 2025 headline, which has no date term to correct.

Sensitivity (Experiment 4). On the center-only grid the anchor rate runs from **0%** (summer solstice only, $\pm$50 years, whether or not the catalog is precision-filtered) to **46%** (all events, summer solstice only, $\pm$400 and $\pm$500 years). Precision-filtered events with summer solstice only give **23%** at $\pm$250 years (3 of 13). Figure 2 uses the precision-filtered labels. Half-widths do not set the match window in this grid.

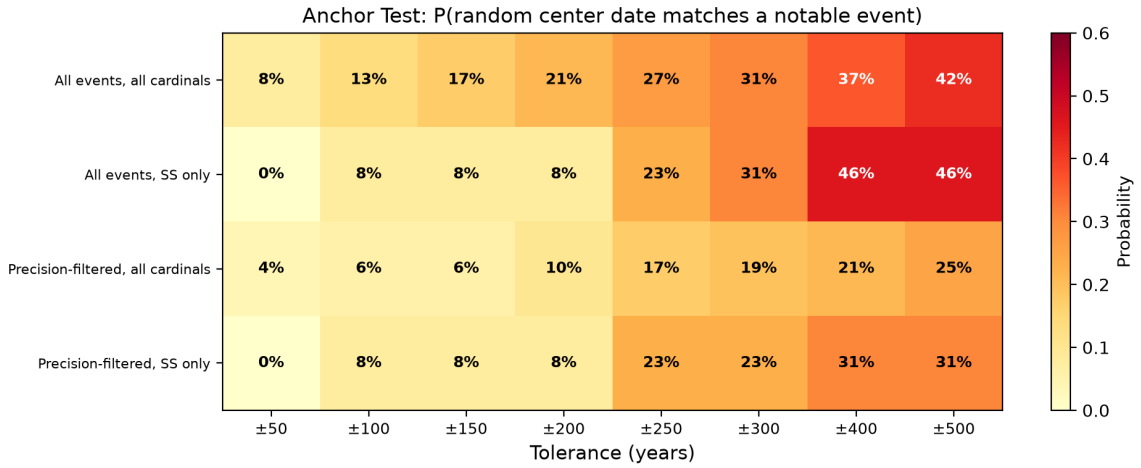


Figure 2. Center-date anchor rate by tolerance and scenario. Rows use the precision-filtered labels.

Multiverse (Experiment 8). All 128 paths have positive joint probability, and the published date is a hit on every one, so no path is excluded. The reference path matches the Younger Dryas cluster. Its joint is 5.90×10^{-7} (1 in 1,693,817).

Across the 32 primary paths the joint runs from 1.48×10^{-7} (1 in 6,775,270: all four cardinal days, precision-filtered catalog, $\pm$100 years, center anchor) to 1.57×10^{-6} (1 in 635,182: summer solstice only, all events, $\pm$100 years, window anchor), a factor of about **11**. The median is 7.13×10^{-7} (1 in 1,401,780). Thirteen of the 32 (40.6%) are at least as small as the reference. That places the reference near the middle of the distribution and says nothing more. Table 5 has the summary, and the full path table is in `output/results.json`.

Adding the 96 paths that drop animals after the ranks are seen widens the range to 7.22×10^{-6} (1 in 138,424), a factor of about **49** overall. Dropping animals is a stronger degree of freedom than choosing center versus window, and those extended paths are not on the same footing as the primary set.

Table 5. Primary multiverse ($n = 32$, all six animals). The reference row is the exploratory product, not $p \times 2/7$.

Statistic	Joint P	≈ 1 in
Minimum (most significant)	1.48×10^{-7}	6,775,270
Median	7.13×10^{-7}	1,401,780
Exploratory reference path	5.90×10^{-7}	1,693,817
Maximum (least significant)	1.57×10^{-6}	635,182
Paths at least as significant as the reference	$13/32 = 40.6\%$	—

Ranking sensitivity (Experiment 12). Adding 1 to every published rank moves the six-animal probability from 1 in 390,881 to 1 in 53,912 (Table 6). Setting the top three ranks to 3 instead of 1 leaves that same six-animal probability and changes the pre-specified top-three piece from 1 in 85,184 to 1 in 1,015. These are hand-built stress tests. They show that the published probability moves when the rank vector moves. They are not a blind replication.

Table 6. Ranking sensitivity.

Scheme	Ranks	Sum	$P(\text{sum} \leq S)$	≈ 1 in
Sweatman original	[1, 1, 1, 4, 5, 6]	18	2.56×10^{-6}	390,881
Conservative (+1 each)	[2, 2, 2, 5, 6, 7]	24	1.85×10^{-5}	53,912
Skeptical (top 3 at rank 3)	[3, 3, 3, 4, 5, 6]	24	1.85×10^{-5}	53,912
Mixed (top 3 vary)	[1, 2, 3, 4, 5, 6]	21	7.48×10^{-6}	133,722
Degraded (top 3 at rank 5)	[5, 5, 5, 4, 5, 6]	30	8.18×10^{-5}	12,221
Generous (bottom 3 improved)	[1, 1, 1, 2, 3, 4]	12	1.27×10^{-7}	7,853,154

5. Discussion

5.1 What the evidence actually supports

Inside the published scoring, three animals are ranked first of 44. If scorpion → Scorpius, canid → Lupus, and bird-with-snake → Ophiuchus were named in advance, three independent rank-1 draws have probability 1 in 85,184. The code, and the prose that follows it, sorts six ranks and keeps the best three. The null for that procedure is 1 in 4,484. Either figure is the core of the visual claim, and they answer different questions. The canid identification is the weakest of the three on archaeological grounds even though its published rank is 1 (Section 5.2).

The date component is a different claim, and under this catalog it is common: 41.8% of the analysis window is already within $\pm$250 years of some listed event, and 26.9% of random anchors match on the center date. It is not part of $p \times 2/7$. The bottom three mappings are significant at a conventional 0.05 threshold and carry much less weight than the top three. The published date is itself a Younger Dryas match, 50 years from the cluster center. What the tests show is that such

a match is cheap. Exploratory joints move with the path. Across the primary set the move is a factor of about 11; once post-hoc animal dropping is included it is a factor of about 49.

5.2 The self-scoring problem

Sweatman and colleagues ranked all $6 \times 44 = 264$ animal–constellation pairs themselves, knowing the hypothesis. The papers give verbal reasons for each chosen match. This is a structural limit, not an accusation about intent. Subjective alignment judgments in archaeoastronomy are a known vulnerability (Ruggles, 2015; Silva, 2020). The people who stated the hypothesis also produced the ranks the test consumes.

The ranks also stack three questions: whether ancient observers named constellations at all, whether they grouped the same stars the later Greek tradition grouped, and whether they used the same animals. This analysis only scores the last question inside the modern Western figures Sweatman adopted. Some matches are helped by the shared animal name (scorpion → Scorpius, canid → Lupus) rather than by a star pattern that would force that reading on its own. Sweatman & Gerogiorgis (2025) call the Scorpius and Lupus ranks objective and flag some uncertainty only for Ophiuchus. The carving's species is still only a canid, and that identification should stay marked tentative until someone who does not know the hypothesis repeats it. A blind ranking would test whether the published similarity judgments replicate. It would not show that Pre-Pottery Neolithic observers partitioned the sky as later Mediterranean traditions did.

5.3 Implications for archaeoastronomy methodology

Date coincidences in the deglacial period are inexpensive. A study that aligns an archaeoastronomical date with a "notable" event should report the coverage of its own window. Without that baseline, a crowded catalog manufactures alignments.

Visual scores should be produced by raters who do not know the hypothesis. The rank-sum machinery is usable. The open problem is the input. Silva (2020) treats structural orientations, not the iconographic comparison that carries Pillar 43.

Analyst choices should be shown as a multiverse, with a validity filter. A path that makes the observed reading a miss is not an alternative evaluation of that reading. The observed reading has to be the date the authors publish, not a stand-in for it such as a constellation midpoint. Zero-probability paths belong in the denominator disclosure even when, as here, there are none.

The event catalog should stop at the monument. Events later than the carving cannot be commemorated by it. Counting them raises the baseline; here it moves coverage from 33.3% to 41.8%.

A pre-specified trio and the best three of six are different tests. Reporting $1/44^3$ for animals chosen after the ranks are known overstates how rare the pattern is.

Leave-one-out on both the events and the animals shows which pieces carry the result. Here no single event cluster carries the date baseline, and three animals carry the visual result.

5.4 Limitations

The core rank tests accept the published ranks. Experiment 12 only perturbs them. The event catalog is ours. The sweep and the circular shift keep the qualitative conclusion, that a date match is not rare, but the percentages in Section 4 are catalog-dependent. In particular:

- The mammoth row is still 7700 BCE with half-width 300 years. That is **not** an IntCal age for Kuzmin & Orlova's ca. 9700 ^{14}C BP mainland extinction, and it is not a calendar reading of the Wrangel 7700 ^{14}C BP survival. Robustness check B shows it does not matter here. The 9,150 BCE stand-in used in that check assumes an error term the source does not give.

- The strict null draws constellation midpoints, while the published date comes from the disc's position inside the constellation. The position null (23.9% at $\pm$250 years) is closer to that procedure. Neither reproduces how Sweatman fixed the date.
- The 9,500 BC cut-off in robustness check A is a round figure inside Enclosure D's calibrated range. The PBO cluster (9,400 BC) falls just after it.
- Ranks 4, 5, and 6 are published as upper bounds and are used as exact.
- Heinrich 1 is placed at Hemming's 16,800 calendar years under the 1950 convention. Later work splits Heinrich layer 1 into more than one peak. The $\pm$500 year half-width does not turn 14,850 BCE into that later chronology.
- Mazama uses Zdanowicz et al. (1999), not a newer Bayesian tephra age.
- Only the Pisces/Aries junction was moved to the IAU-boundary intersection. The other boundaries are a one-decimal table.
- No Stellarium ephemeris was compared with the linear engine beyond the two internal longitude checks in Section 3.1.
- Thirty of 52 center dates lie in the 20,000–5,000 BC window. The anchor probability over 52 is not a probability for a year drawn uniformly from that window. Coverage is that second probability, at 41.8% for $\pm$250 years, and the strict rate among the 30 in-window anchors is 46.7%.
- Three cluster pairs closer than 500 years were left unmerged. Independence of the 14 clusters is an analyst choice.

The linear precession approximation is about 1° over 13,000 years, small next to a typical ecliptic constellation and worth about 70 years. Dates in this paper are not good to a decade.

The 2024 lunisolar-calendar reading (Sweatman, 2024) is not tested. If it holds, it is a separate line of evidence.

Archaeological objections (whether the constellations could have been known, whether the pillars are in place, whether the radiocarbon ranges fit 10,950 BC) are those of Notroff et al. (2017) and the excavation reports. They are not answered by the p-values above.

5.5 The blind ranking experiment

The open question is whether the three strong visual matches are forced by the figures or supplied by the hypothesis. A usable protocol is: standardized images of the six carvings and of the 44 constellation figures; raters with no knowledge of Göbekli Tepe, archaeoastronomy, or the Sweatman hypothesis; a preregistered score. The outcome that matters is whether scorpion, canid, and bird-with-snake are still rank 1 for Scorpius, Lupus, and Ophiuchus when that context is absent.

If those three ranks replicate, the visual signal is at least reproducible inside the modern figure set, and the pre-specified calculation becomes the relevant one only if the trio was locked before the ranks were seen. If they do not replicate, the reported rank-sum rests on judgments that were not separated from the desired outcome. The experiment would not decide whether Pre-Pottery Neolithic observers used those groupings.

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6. Conclusion

The 2025 headline of about 1 in 1.4 million is the rank-sum probability times **2/7**. Our evaluation of that product is 1 in 1,368,083. It does not contain a date. Three further points follow from the recomputed tests.

First, a date match is not rare on this catalog. The published date does land on the Younger Dryas cluster, 50 years from its center. But **41.8%** of 20,000–5,000 BC lies within ±250 years of a listed event, **26.9%** of random anchors match at the center date, and **57.7%** match somewhere in the constellation window. Counting only the events that predate the monument, coverage is still **33.3%**.

Second, the visual weight sits in three of the six mappings. As a pre-specified trio that is 1 in 85,184. As the best three of six it is 1 in 4,484. The other three are 1 in 187. Those probabilities do not multiply to the six-animal result.

Third, the exploratory joint is a distribution. Across the 32 primary paths the joints span a factor of about 11 (1.48×10^{-7} to 1.57×10^{-6}). Including post-hoc animal dropping widens the span to a factor of about 49.

Until a blind ranking is done, the Pillar 43 claim is an unvalidated hypothesis. Its published 1-in-1.4-million figure should not be read as a rare date coincidence multiplied by a rare visual match.

AI Disclosure

This research was conducted using Claude (Anthropic) as a technical reasoning partner for statistical design, code development, and manuscript drafting. ChatGPT (OpenAI) served as an independent reviewer of the statistical methodology and code across multiple review cycles. Accuracy reviews informing versions 2 and 3 were performed with Grok, Codex, and Claude. All numbers reported here were recomputed from `gobekli_statistical_test.py`. The analysis was executed on an NVIDIA DGX Spark. The author takes full responsibility for the scientific content and conclusions.

Data Availability

Script, run log, results JSON, and figures:

`gobekli_statistical_test.py`
`output/run_log.txt`
`output/results.json`
`output/fig1_timeline.png` ... `output/fig5_catalog_sweep.png`

Repository: https://github.com/simossss/gobekli-pillar43-statistics

Zenodo, all versions: https://doi.org/10.5281/zenodo.20364946

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Supplementary materials

- S1. `output/run_log.txt`: console log of this run.
- S2. `output/results.json`: numerical output, including every multiverse path and its `observed_hit` flag.
- S3. Primary multiverse paths: the 32 rows in S2 with `animals` = "All 6 (sum=18)".
- S4. Leave-one-event-out: `results.json` → `leave_one_out`.
- S5. Leave-one-animal-out: `results.json` → `leave_one_animal_out`.
- S6. Ranking sensitivity: `results.json` → `ranking_sensitivity`.
- S7. Selection-adjusted top-3: `decomposition.top3_selection_adjusted_p` in S2, derived in Section 4.2.
- S8. Stellarium residual table: **not run**. Do not treat Section 3.1's two longitude checks as that table.
- S9. Robustness check A: `results.json` → `pre_monument`.
- S10. Robustness check B: `results.json` → `mammoth_sensitivity`.

Appendix A. Changes since earlier versions

A.1 Version 3 against version 2

Version 2 is 10.5281/zenodo.23593750. Its PDF was a draft build and carries draft notices on its first and last pages.

#	Change
1	**The observed reading is the published date.** Version 2 tested the Sagittarius midpoint (11,835 BC) in place of Sweatman's date (10,950 BC). It therefore said the strict hit was an Older Dryas hit and not a Younger Dryas hit. That statement is withdrawn. The published date is 50 years from the Younger Dryas cluster.
2	**Multiverse.** With the published date as the observation, no path is a miss. Primary paths: 32 valid (version 2: 24). Share at least as significant as the reference: 13/32 = 40.6% (version 2: 5/24 = 20.8%). Span: a factor of about 11 (version 2: 3.6). The share is no longer reported in the abstract.
3	**New: events that predate the monument.** Coverage 33.3%, strict 7/52, broad 21/52 at $\pm$250 years.
4	**New: mammoth cluster check.** Removing or recalibrating the placeholder leaves the strict rate at 14/52 or 15/52.
5	**New: in-window strict rate** (14/30 = 46.7%) and **position null** (23.9%).
6	Sweatman & Gerogiorgis (2025) details added: ranks 4–6 are upper bounds, the date is quoted as $\pm$200 years, the Greek-set sum is 26, and the authors' own uncertainty is on Ophiuchus.
7	Figure 1 redrawn so event labels do not overlap. Version history moved out of the body into this appendix.

Unchanged from version 2: the rank-sum, the 2/7 product, the decomposition, both leave-one-out tables, the anchor rates, coverage, the circular shift, and the catalog sweep.

A.2 Version 2 against version 1

Version 1 is 10.5281/zenodo.20364947.

#	Severity	Change
1	**Critical**	Sweatman & Gerogiorgis (2025) **1 in 1.4 million** is rank-sum probability × **2/7** (top-row reversal × spatial correlation). It has **no date term**. Date coincidence is a separate exploratory test. Their own sensitivity range, switching stick-figure sets, is about **1.4 million to 110 thousand**. Our exact product is 7.31×10^{-7} (**1 in 1,368,083**).
2	**High**	Multiverse **validity filter**: the observed Sagittarius × summer-solstice reading must be a hit. Version 2: **5/24 = 20.8%** of valid primary paths are as or more significant than the exploratory reference path. **0** paths have joint probability 0. **8/32** primary paths are observed misses and are excluded from that percentage. version 1 reported **42%** after dropping zero-P paths and counting misses. *Superseded in version 3: the filter tested the Sagittarius midpoint, not the published date.*
3	**High**	Top-3 **1 in 85,184** is only the pre-specified trio ($1/44^3$). Sort-then-pick gives $P(\geq 3 \text{ of } 6 \text{ ranks} = 1) = 2.230 \times 10^{-4}$ (**1 in 4,484**). $P(\text{top}) \times P(\text{bottom}) \neq P(\text{all six})$.
4	**High**	`constellation_date_range()` no longer folds endpoints across the precession branch cut. On the **unchanged version 1 catalog** that fix moves the broad match from **35/52 (67.3%)** to **31/52 (59.6%)**. After the catalog and Pisces/Aries updates below, version 2's broad match is **30/52 (57.7%)** and the strict match is **14/52 (26.9%)**.
5	**High**	Heinrich 1 set to **14,850 BCE** (Hemming 2004: 16,800 calendar years, 1950 BP convention). Mazama set to **5,677 $\pm$ 150 BCE** (Zdanowicz et al. 1999). Mammoth **7,700 BCE kept as an uncalibrated placeholder**: Kuzmin & Orlova (2004) publish ^{14}C ages, not a calendar date. "Major-only" renamed **precision-filtered** (half-width $\leq$ 250 yr). Matching uses the stated tolerance, not half-width.

#	Severity	Change
6	**High**	§2.3: 2017 ≈ 100 million → 2019 ≈ **140 million** is a **stronger** claim, not a reduction. The 2017 statistic was not summed ranks. The 2019 appendix includes a **×13** factor that treats scorpion = Scorpius as given.
7	**Medium+**	Equinox prose: **Virgo = spring**, **Pisces = autumn**. The pool of 44 is Ptolemy's 48 minus Canis Major, Lepus, Eridanus, and Argo Navis, not every constellation visible from the site. Notroff's objection is paraphrased, not quoted as "incredibly arbitrary." Canid → Lupus marked tentative. Dietrich et al. (2013) cited as calibrated ranges for Enclosure D, not a site-wide Gaussian. Coverage is **41.8%** in version 2 (version 1 results were 42.5%; the version 1 abstract's 43% rounded that figure).
8	**Code**	Pisces/Aries ecliptic junction set to **28.687°** (IAU boundary ∩ J2000 ecliptic; the review's ~28.8°). Other junctions stay at the version 1 one-decimal table.
9	**Citations**	Sweatman 2025 URL and not-peer-reviewed status; Sweatman 2024 full cite and DOI; Ruggles as editor; Ryan et al.; Holliday et al. 2023 as optional YDIH context.